

is to be able to identify, isolate and analyse a system as quickly and efficiently as possible. Having an effective touchscreen implementation on an instrument can reduce the time it takes to configure the instrument and analyse signals," explains Tangri.

Quality brings success

The definition of quality can be subjective, often determined by the customer's unique perspective. Nevertheless, experts who provided inputs for this report believe that objective standards of quality contribute towards quantifiable benchmarks. Narendra Goliya says, "As every customer segment wants its systems to function properly and to ensure reliability on consistent basis, focus on quality and safety is increasing gradually. Customers are willing to spend extra money for good quality."

Quality parameters can be divided into three categories:

1. Functional
2. Safety
3. Reliability

"Separate focus on safety would be necessary in times to come and India must have a special safety certification lab and adherence to IEC norms. At the same time we must play an active role in the formation of IEC standards," opines Narendra Goliya.

The following quality and safety norms would always remain important in this segment:

1. Compliance to over-voltage

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—Premchand Goliya, chairman and managing director, Meco Instruments Pvt Ltd



categories like CAT IV, III, II and I that ensure instruments to be safe in different working environments

2. Compliance to Ingress Protection that ensures protection from dust and water
 3. Compliance to IEC 61010-1 Safety Norms for T&M products
- Goyal comments, "Testing can be extremely hazardous, if not done in the right way with the right equipment. One of the core values of my organisation is safety. Testing equipment must comply with CAT, IEC, IEEE, CIGRE and other such norms."

Narendra Goliya feels that enforcement of quality standards (for example, harmonics levels) should be mandated by the government. He says, "Time has come to revive important quality norms like ISI. Non-adherence to standard engineering and commercial practices by various unscrupulous players are the further roadblocks in the growth of this sector."

"Quality takes a backseat when government tender business is done only on L1 basis. It would be essential for the government to foster quality practices and build a quality culture by adding screening on quality and specification front before a tender is floated."

Quality requirements in end-use segments can also accelerate growth of the T&M industry. However, bringing in increased focus on quality in end-use segments can be time consuming. This slow pace of change will no doubt hinder growth of the T&M sector.

For example, Tripathi says, "The telecom sector has invested hugely in spectrum procurement but is not willing to invest in T&M. Rather, mobile network operators have adopted rental model, managed services model and CAPEX model that reduce procurement of T&M equipment in this segment."

Other industry experts also feel

"Connected lifestyles, M2M, smartcities and the IoT are some of the trends that will lead to the growth of the T&M industry. Connected cars, connected

lifestyles, higher data and lower cost of smartphones are some trends seen in the telecom market. Operators are using new technologies like MIMO, Vo LTE, Advanced and LTE Carrier Aggregation to improve spectrum efficiency and customer experience of high data speed."

—Madhukar Tripathi, senior manager-marketing, Anritsu India Pvt Ltd



the importance of making customers aware and convinced about quality, safety and maintenance practices through the use of right T&M instruments. They expect positive intervention of regulatory bodies to facilitate this shift.

"BEE should implement energy-efficiency requirement more vigorously in local production of electronic appliances and goods, which will help increase demand for instruments. The government should fund small and medium industries with special grants, say, 50 per cent of the cost for buying instruments for improving quality of production," suggests Premchand Goliya.

Way forward

T&M companies work as solution partners with decision makers in the electronics system design and manufacturing industry by offering solutions and services for leading-edge electronic design and test. The move of ESDM industry players towards adoption of T&M solutions will help move the technology forward.

As Tangri rightly comments, "We have helped many of the world's greatest visionaries and innovators achieve elusive breakthroughs that advance technology. The result? Ideas that are already making the world a better place and technologies that are making businesses more productive." **EFY**